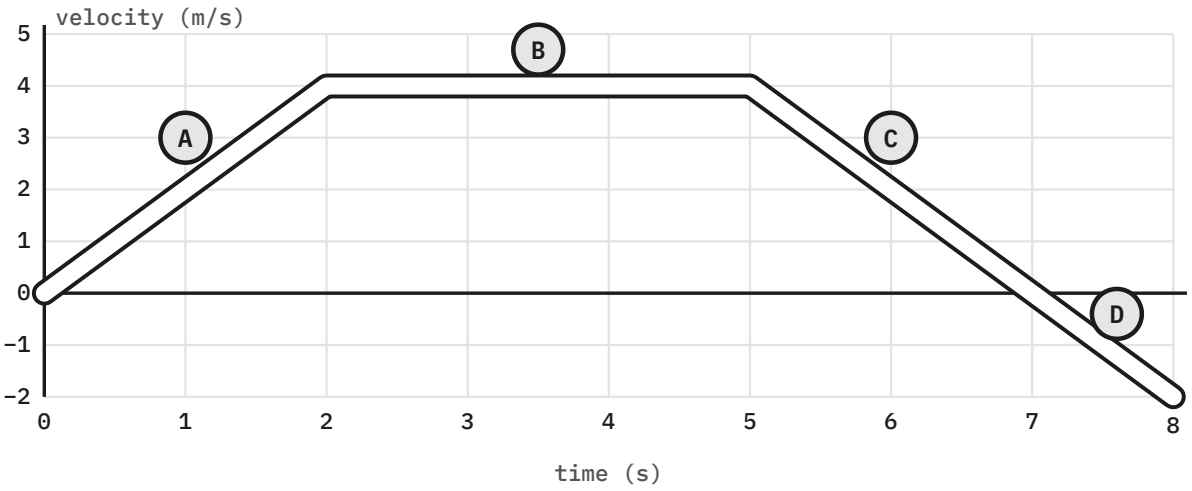
Read every part of the v–t graph

04 F

DIRECTIONS · Vertical axis = velocity. Area between the line and zero = displacement.



A · 0–2 s

speeding / steady / slowing

area = ____ m

B · 2–5 s

v = ____ m/s

area = ____ m

C · 5–7 s

v drops to ____

area = ____ m

D · 7–8 s

v ____ 0 · moving which way?

SPLIT THE ARROW

A velocity of 10 m/s points 37° above +x. v_x = 10 cos 37° = ____ m/s · v_y = 10 sin 37° = ____ m/s

TELL THE MOTION STORY

R2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 4 · R2 · CHOICE BOARD

v–t and vectors tic-tac-toe

04 D

DIRECTIONS · Complete three in a row. Shade each square you finish.

CALCULATE v = +3 m/s for 4 s. Shade the area and find Δx.	**DRAW** Sketch a v–t graph for a bike that rides steadily, then stops.	**EXPLAIN** Why can a flat v–t line mean the object is moving?
REAL WORLD A plane flies 200 m/s at 30° above level. Find its vertical part.	**CHALLENGE ★** v = −2 m/s for 3 s, then +4 m/s for 2 s. Total Δx?	**CALCULATE** Split 20 m/s at 53° from +x into parts.
EXPLAIN Why must you say which axis an angle starts from?	**DRAW** Draw a 5 m/s arrow at 37° and its two parts.	**REAL WORLD** A boat heads east at 4 m/s; a current pushes north at 3 m/s. One-arrow speed?

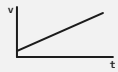
Three in a row: _____ · The square that taught me most: _____

R2 1 2 3

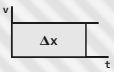
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it: graphs and arrows

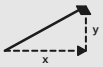
Language goal: I can **describe** a velocity–time graph.



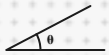
v–t graph
gráfica v–t · gráfico v–t
velocity on the up axis



area
área · área
height × width = Δx



component
componente · componente
one part of an arrow



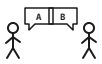
angle
ángulo · ângulo
measured from +x

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: flat · area · displacement · component · angle

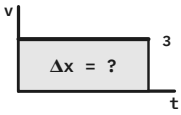
On a v–t graph, a ____ line means steady velocity. The ____ under the line is the ____. An arrow can be split into an x ____ and a y part. We measure the ____ from the +x axis.

2 · SAY IT TO A PARTNER, THEN WRITE IT



The line is flat at ____ m/s, so the object moves at a ____ velocity.
The area is ____ × ____ = ____ m, so the displacement is ____ m.

3 · SOLVE WITH THE STEPS



v = +3 m/s for 4 s on a v–t graph.

STEP 1 · HEIGHT AND WIDTH	STEP 2 · MULTIPLY	STEP 3 · SAY IT
height ____ m/s · width ____ s	____ × ____ = ____ m	"The displacement is ____ m."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ A flat line does not mean stopped because ____ .
☐ The area tells me ____ because ____ .
☐ The x part is bigger than the y part because ____ .

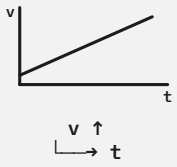
I said it ☐ · I wrote it ☐

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Picture dictionary: graphs + vectors

Language goal: I can **describe** a graph and an arrow.



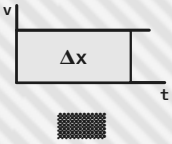
v–t graph

gráfica v–t · gráfico v–t

velocity on the up axis

my language: _____

use it: _____



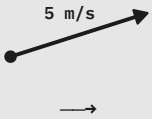
area

área · área

height × width = Δx

my language: _____

use it: _____



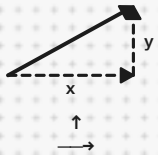
vector

vector · vetor

size + direction

my language: _____

use it: _____



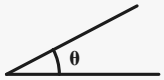
component

componente · componente

one part of an arrow

my language: _____

use it: _____



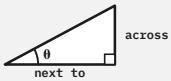
angle θ

ángulo · ângulo

measured from +x

my language: _____

use it: _____



next to
across

cos / sin

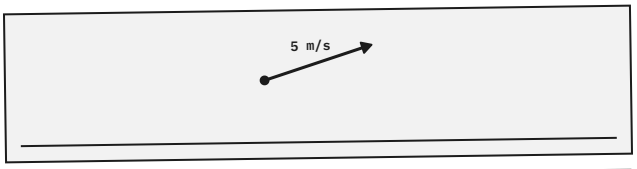
coseno/seno · cosseno/seno

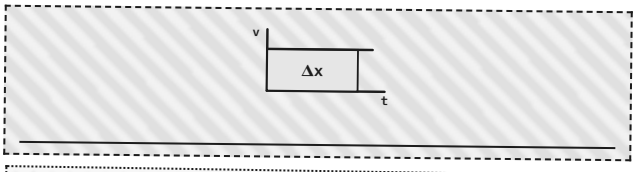
pick the right part

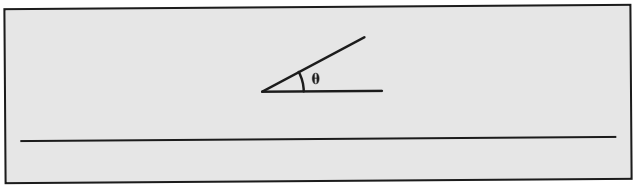
my language: _____

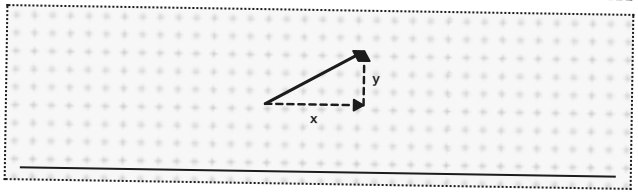
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

ANTOCCI ENGINEERING · DWG U1-04 · R2

TRACKING DISH: AIM AT 2026-XJ

Story velocity: 8,500 m/s at 15° from the toward-Earth (+x) axis.

CALC 1 · TOWARD-EARTH PART

$v_x = 8{,}500 \times \cos 15° =$ ____ m/s

CALC 2 · SIDEWAYS PART

$v_y = 8{,}500 \times \sin 15° =$ ____ m/s

DRAWING · SKETCH THE ARROW AND ITS TWO PARTS (label the angle and axis)

CALC 3 · CHECK

Does $\sqrt{(v_x^2 + v_y^2)}$ give back 8,500? Show it.

CALC 4 · WHICH IS BIGGER?

Why is the sideways part small but still important?

ENGINEER'S NOTE

Write the aiming instructions for the dish crew.

R2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 4 · R2 · LOOP CHAIN

I have... who has?

04 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, draw an arrow, repeat.

★ START — **A**

WHO HAS…

v = +3 m/s for 4 s. What is Δx?

I have **8 m/s** — **B**

WHO HAS…

10 m/s at 37°: v_y = 10 sin 37°?

I have **stopped** — **C**

WHO HAS…

v = −2 m/s for 3 s. What is Δx?

I have **5** — **D**

WHO HAS…

The area between a v–t line and zero is…?

I have **+12 m** — **E**

WHO HAS…

A flat v–t line at 0. What is the object doing?

I have **displacement** — **F**

WHO HAS…

A flat v–t line at +5 m/s means…?

I have **−6 m** — **G**

WHO HAS…

10 m/s at 37°: v_x = 10 cos 37°?

I have **6 m/s** — **H**

WHO HAS…

Parts 3 east, 4 north. Size of the arrow?

I have **steady velocity** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

R2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

UNIT 1 · DAY 4 · R2 · COMIC STRIP

Comic: a flat line still moves

04 CS

YOUR READER
A 6th grader who thinks a flat v–t graph line means "not moving."

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **PURPLE** — Same-length velocity arrows on the cruising car
- ☐ ◌ **YELLOW** — The flat line for the steady car
- ☐ ◯ **PENCIL** — Both axes labeled **v (m/s)** and **t (s)**

1 · CRUISING *A car driving at a steady 10 m/s.*

CAPTION _____

2 · THE GRAPH *A character points to the flat line on a v–t graph.*

CAPTION _____

3 · THE TWIST *A parked car. Show its v–t line, too.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **velocity · constant · v–t graph · at rest**

R2 1 2 3

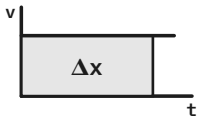
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · signed area

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Signed area · área con signo (ES) · área com sinal (PT)



TODAY'S CASE

A cart moves at +2 m/s for 3 s and then at −2 m/s for 1 s.

01 · SKETCH AND LABEL

Draw both intervals on a velocity–time graph. Hatch the area below zero differently.

02 · CHANGE THE PICTURE

Label the displacement for each interval, including its sign. Draw one blue arrow showing the combined displacement.

Annotate your sketches. Circle the mark that best shows signed area. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The area below zero counts as …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · signed area

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Signed area · área con signo (ES) · área com sinal (PT)

A · An area below the time axis represents negative displacement.

B · Adding the magnitudes of both areas gives displacement in every trip.

01 · DEFINITION

What does signed area mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else